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ORIGINAL ARTICLE

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Economic Hardship, Ocular Complications, and Poor Self-reported Visual Function are Predictors of Mental Problems in Patients with Uveitis

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ABSTRACT

Purpose: To characterize the quality of life and mental health status of patients with uveitis and investigate predictors of psychological problems.

Methods: A total of 245 patients and 105 controls were enrolled in this cross-sectional study. Quality of life, psychological status, socio-demographic and clinical data were obtained from questionnaires and medical records. Multivariate regression analyses and Receiver Operating Characteristic (ROC) were applied to obtain the model predicting psychological problems of patients.

Results: Of 245 patients, 16.7% and 26.5% ($P < .0001$) screened positive for anxiety and depression, respectively. The model predicting anxiety was comprised of low annual household income and poor self-reported visual function ($P = .029$, $P < .0001$, respectively), with an AUC of ROC of 0.744. The model predicting depression was comprised of poor self-reported visual function and ocular complications ($P < .0001$, $P = .012$, respectively), with an AUC of 0.78.

Conclusions: Economic hardship, ocular complications, and poor self-reported visual function are predictors of mental problems in patients with uveitis.

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Quality of life; mental health; depression; anxiety; uveitis; predictors

Uveitis is a potentially blinding inflammatory disease causing 5–10% of visual impairment worldwide[1] with up to 35% of patients with uveitis suffering from severe vision loss, including legal blindness.[2,3] The prevalence of uveitis in the Chinese population was about 1.52‰, the blind rate is about 0.86‰,[4] and is on the rise.[5] As an inflammatory disease, uveitis can also lead to psychological problems as well,[2,6,7] such as anxiety and depression.[8,9]

Based on the bio-psycho-social medical model, the connections between psychological, physical, and social factors must be considered when assessing their influence on patients. For example, more negative socio-demographic characteristics may increase the occurrence of psychological problems[10,11] and clinical features, like poor visual function, are also risk factors for psychological problems.[12] Studies have shown psychological problems can affect patient treatment compliance and ultimately affect the efficacy of treatment.[13,14] Early detection of and timely intervention for psychological problems may prevent the occurrence of adverse events and improve patient prognosis. In addition, prediction models for psychological problems may provide clinicians with some guidance in terms of drug selection, surgical timing, disease education, and clinical care, as well as prompt the development of medical security for patients with uveitis.

With the help of self-administered questionnaires, patients with retinitis pigmentosa,[15] macular degeneration,[16] and newly diagnosed glaucoma[17,18] were found to display significant depressive symptoms and poor self-reported health and vision. At present, while there have been studies abroad focusing on the mental health and quality of life of patients with uveitis,[19–21] there has been almost no domestic research.

The purpose of this study was to assess the mental health status and vision- and health-related quality of life of patients with uveitis in China and identify risk factors related to mental health. To our knowledge, this study is the first to focus on predictors of psychological problems in Chinese patients with uveitis.

Materials and Methods

Study Subjects

A cross-sectional study design was employed. The enrolled patients were treated at the Uveitis Clinic at the Eye Hospital of Wenzhou Medical University in China between July 2018 and February 2019. The eligible individuals were Chinese-

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speaking patients, who were at least 18 years old with a well-defined diagnosis of infectious or noninfectious uveitis. Patients diagnosed with severe and active systemic disease unrelated to uveitis were not included in this study. In addition, patients with a clearly diagnosed history of mental and neurological diseases, alcohol dependence, drug dependence unrelated to the disease, or hearing impairment or who refused to participate or did not cooperate were not included in this study. Participants who had no prior history of eye disease that affected vision and within the patient distribution in age, sex, income, and education were enrolled in the reference group.

Based on experience, the sample size for the multivariate logistics regression analysis should be 10–20-fold that of the research factors. Since there were approximately 10 relevant factors in our experimental group, the sample size of the experimental group was set as 200. Considering the integrity of the data, 240 patients were expected to be enrolled in the experimental group. In addition, about 100 eligible people with no history of eye disease were expected to be enrolled in the control group. This study was approved by the review committee of Wenzhou Medical University and is in line with the Helsinki declaration.

Design and Procedures

To ensure the reliability and professionalism of diagnoses and avoid any possible influence on the reliability of the experimental data due to patient doubt, uveitis specialists with medical qualifications were required to provide treatment to the patients.

Before data collection, the purpose of the study and ethical questions was clearly explained to the subjects and questions related to the study were answered. After obtaining trust and informed consent from the subjects, the collection of relevant data formally began. Subjects were asked to complete three self-administered questionnaires related to the quality of life (QOL) and mental health, as well as a questionnaire on their demographic characteristics by themselves unless they were illiterate or unable to do so due to poor vision. In these cases, a trained interviewer read and completed the questionnaires based on patient answers. After completion of the questionnaires, the patients were given a physical ophthalmic examination and the necessary clinical and auxiliary examinations required by the doctor according to their own conditions.

Socio-Demographics and Clinical Characteristics

The socio-demographic characteristics of the subjects collected with the questionnaires included gender, age, nationality, education, occupation, and annual household income. The clinical characteristics of the patients were collected from medical records and included laterality of disease, current best-corrected visual acuity (BCVA), duration of disease, course of disease (acute, recurrent, or chronic), diagnosis of diseases, anatomic location of inflammation, uveitis-associated systemic inflammatory diseases, current activity of disease, current and previous treatments, history of changing immunomodulatory therapy (IMT), current ocular comorbidities, previous and current topical injection therapy, and previous ocular surgery. The BCVA was recorded individually for each eye and then converted to the logarithm of the minimal angle of resolution (logMAR) scale. Low vision based on finger counting, hand motion, light perception, and no light perception was recorded as a logMAR 1.7, 1.8, 1.9, and 2.0, respectively.[22] The eyes of each patient were designated as better or worse according to their logMAR visual acuity score. The course, anatomic location of inflammation, and the grading of anterior chamber cells were recorded as outlined by the Standardization of Uveitis Nomenclature criteria.[23] Active uveitis was defined as active inflammatory; retinal vasculitis, retinitis, choroiditis, cystoid macular edema, or papillitis[9]; vitreous haze grade of 1+ or higher[9,23,24]; or anterior chamber cell grade of 1+ or higher.[9,23] As for Posner–Schlossman syndrome, the presence of transient episode of elevated intraocular pressure or mild anterior chamber inflammation can be defined as activite uveitis.[23,25] Patients with Fuchs' syndrome were classified as having active uveitis because they had chronic anterior and vitreous inflammation.

Quality of Life and Mental Health

Vision-related quality of life (VRQOL), health-related quality of life (HRQOL), and mental health data were collected through self-administered questionnaires, including a Chinese version of the 25-item National Eye Institute Visual Function Questionnaire (CHI-VFQ-25), a Chinese version of the 36-item Short Form Health Survey (CHI-SF-36), and a Chinese version of the Hospital Anxiety and Depression Scale (CHI-HADS). The cutoff values for the HADS-Anxiety (HADS-A) and HADS-Depression (HADS-D), which were used to evaluate the positive tendencies for anxiety and depression in the subjects, were set as 8. Please see supplementary information for a detailed description of the questionnaires.

Statistical Analyses

Statistical analyses were conducted using SPSS statistical software (SPSS Inc, version 14.0, Chicago, Illinois). Wilcoxon rank-sum tests were conducted to compare the scores of self-administered questionnaires between patients and controls. The Chi-square test was conducted to determine differences in categorical variables between the anxious and not anxious groups, as well as between the depressed and not depressed groups. Continuous variables were analyzed using Wilcoxon rank-sum tests or Student's t-tests. Associations between HADS-A or HADS-D scores and VFQ-25 composite scores were measured using Spearman correlation coefficient. Univariate analyses were conducted to compare sociodemographic and clinical characteristics between patients with and without a positive screen for anxiety or depression. All tests described above were two-sided and *P*-value less than 0.05 was set as the significant level. A forward stepwise multivariate regression model was used to determine significant predictors for anxiety and depression with an entry *P*-value

less than 0.10 and a removal *P*-value more than or equal to 0.15. Additionally, receiver operating characteristic (ROC) curves and area under the curve (AUC) were performed to determine the best model predicting anxiety and depression.

Results

QOL and Mental Health Status of Uveitis Patients

A total of 260 patients and 115 reference subjects with similar age, sex, income, and education distributions were asked to participate in this study. Of these, 245 patients (94.23%) and 105 reference subjects (91.3%) completely answered the questionnaires. Scores from the patient and reference groups for the three scales are presented in Supplementary Table S1. The patients with uveitis scored significantly lower on the CHI-VFQ-25 and CHI-SF-36 for nearly all dimensions. For the HADS-D subscale, 65 (26.5%) of the patients with uveitis had scores of at least 8 compared to the only 10 (9.5%) in the reference group, and the patients with uveitis scored significantly higher than the reference group ($P < .0001$).

Socio-Demographics and Clinical Data

Based on 8 as the cutoff value of HADS to evaluate the positive tendency for anxiety and depression in the subjects, 41 (16.7%) patients were considered "anxious" and 204 (83.3%) were considered "not anxious." Similarly, 65 (26.5%) patients were considered "depressed," while 180 (73.5%) patients were considered "not depressed" (Table 1).

Overall, 49% of the patients were men and the mean age of the patients was 42 years. The anxious and depressed groups had a higher percentage of people with lower education levels than the not anxious and not depressed groups ($P = .049$ and $P = .005$, respectively). Compared with 12 (5.9%) participants in the not anxious group, 8 participants (19.5%) in the anxious group discontinued work due to uveitis ($P = .009$). The percentage of patients with an annual household income lower than 100,000￥ was significantly greater in the anxious and depressed groups than in the not anxious and not depressed groups (78.0% vs 53.9%, $P = .004$ and 72.3% vs 52.8%, $P = .006$, respectively).

Systematic prednisolone was administered to 26.5% of patients. To control the disease, 21.2% of patients were taking IMT. Additionally, 15 (23.1%) patients in the depressed group and 22 (12.2%) in the not depressed group had a history of changing their IMT due to an intolerance to the medications or inability to control inflammation ($P = .036$) (Table 2).

At the time of the survey, 41.6% of the patients had suffered from ocular complications caused by uveitis with the majority of cases experiencing complicated cataract, secondary glaucoma, and macular edema (20.8%, 12.7%, and 10.2%, respectively). Notably, there were significantly ($P < .0001$) more patients with complications in the depressed group (39, 60%) than in the not depressed group (63, 35%). The incidence of complicated cataracts and macular edema was also higher (30.8 vs 17.2%, $P = .021$ and 16.9 vs 7.8%, $P = .037$). Furthermore, the incidences of secondary glaucoma in the anxious and depressed groups were significantly higher than in the reference group (24.4 vs 10.3%, $P = .013$ and 21.5

Table 1. Socio-demographic characteristics of 245 patients with uveitis.

		Patients with uveitis grouped according to HADS-A score[a]			Patients with uveitis grouped according to HADS-D score[a]		
Variable	Total (N = 245) median (IQR) mean (SD) n (%)	Anxious (N = 41) median (IQR) mean (SD) n (%)	Not anxious (N = 204) median (IQR) mean (SD) n (%)	*P* value[b]	Depressed (N = 65) median (IQR) mean (SD) n (%)	Not depressed (N = 180) median (IQR) mean (SD) n (%)	*P* value[b]
Male gender, (%)	120 (49.0%)	23 (56.1%)	97 (47.5%)	.318	31 (47.7%)	89 (49.4%)	.809
Age (years), mean (SD)	42 (13)	43 (12)	42 (13)	.395	46 (14)	41 (13)	.453 [c]
Nationality, (%)				.440			.633
Han	240 (98%)	40 (97.6%)	200 (98.0%)		64 (98.5%)	176 (97.8%)	
Bouyei	1 (0.4%)	0 (0%)	1 (0.5%)		1 (1.5%)	0 (0%)	
Yi	1 (0.4%)	1 (2.4%)	0 (0%)		0 (0%)	1 (0.6%)	
She	2 (0.8%)	0 (0%)	2 (1%)		0 (0%)	2 (1.1%)	
Hui	1 (0.4%)	0 (0%)	1 (0.5%)		0 (0%)	1 (0.6%)	
Educational level, (%)				.049			.005
Primary school or less	59 (24.1%)	16 (39.0%)	43 (21.1%)		25 (38.5%)	34 (18.9%)	
Secondary school to high school	123 (50.2%)	17 (41.5%)	106 (52.0%)		29 (44.6%)	94 (52.2%)	
Bachelor's degree and higher	63 (25.7%)	8 (19.5%)	55 (27.0%)		11 (16.9%)	52 (28.9%)	
Occupation, (%)							
Discontinued work due to uveitis	20 (8.2%)	8 (19.5%)	12 (5.9%)	.009	9 (13.8%)	11 (6.1%)	.051
Housewife	33 (13.5%)	4 (9.8%)	29 (14.2%)		7 (10.8%)	26 (14.4%)	
Individual household	58 (23.7%)	12 (29.3%)	46 (22.5%)		15 (23.1%)	43 (23.9%)	
Farmer	5 (2.0%)	1 (2.4%)	4 (2.0%)		2 (3.1%)	3 (1.7%)	
Office worker	89 (36.3%)	13 (31.7%)	76 (37.3%)		20 (30.8%)	69 (38.3%)	
Student	10 (4.1%)	0 (0%)	10 (4.9%)		0 (0%)	10 (5.6%)	
Retiree	17 (6.9%)	1 (2.4%)	16 (7.8%)		8 (12.3%)	9 (5.0%)	
Other	13 (5.3%)	2 (4.9%)	11 (5.4%)		4 (6.2%)	9 (5.0%)	
Annual household income, (%)				.004			.006
≤100000	142 (58.0%)	32 (78.0%)	110 (53.9%)		47 (72.3%)	95 (52.8%)	
>100000	103 (42.0%)	9 (22.0%)	94 (46.1%)		18 (27.7%)	85 (47.2%)	

[a]Chinese version of the Hospital Anxiety and Depression Scale: patients in the anxious and depressed group scored ≥8 and patients in the not anxious and not depressed group scored <8.
[b]Analyzed using a two-sided Chi-square test at a confidence level of 95%.
[c]Analyzed using a two-sided t-test at a confidence level of 95%.

vs 9.4%, $P = .012$, respectively). The median (IQR) number of periocular or intraocular injections in patients in the depressed and not depressed groups were 1 (0–3) and 0 (0–2), respectively ($P = .005$).

The logMAR vision scores in both the better and worse eye in the anxious and depressed groups were significantly worse than in the not anxious and not depressed groups (Table 2). In addition, nearly all the subscale and VFQ-25 composite scores were significantly lower in the anxious and depressed groups (Supplementary Table S2). There was a correlation between the HADS-A scores and VFQ composite scores (Spearman correlation of −0.367, $P < .0001$; Figure 1a) and between the HADS-D scores and VFQ composite scores (Spearman correlation of −0.482, $P < .0001$; Figure 1b).

The QOL of uveitis-related conditions was investigated. Sjögren syndrome associated uveitis and Fuchs' recorded best VRQOL, while sympathetic ophthalmia and Behcet's disease (BD) recorded worst VRQOL (Supplementary Table S3). Fuchs', sarcoidosis, and Possner–Schlossman syndrome (PSS) recorded best physical HRQOL, while sympathetic ophthalmia, Vogt–Koyanagi–Harada syndrome, and BD recorded worst physical HRQOL. Furthermore, Fuchs', HLA-B27-related acute anterior uveitis, and PSS recorded best mental HRQOL, while sarcoidosis, sympathetic ophthalmia, and BD recorded worst mental HRQOL.

Univariate Analysis

A correlation matrix involving all variables predicting positive-HADA and positive-HADD was developed using univariate analysis (Table 3). Patients who discontinued work due to uveitis had a greater probability of being anxious or depressed ($P = .006$ and $P = .057$, respectively). By contrast, a higher education level or annual household income was protective against anxiety ($P = .037$ and $P = .006$, respectively) and depression ($P = .002$ and $P = .007$, respectively). In terms of the clinical traits, patients with worse logMAR vision in either the better and worse-seeing eye were significantly more likely to be anxious and depressed. Ocular complication was also a risk factor for depression (OR = 2.786, 95% CI 1.555–4.991, $P = .001$), among which complicated cataracts ($P = .023$), secondary glaucoma ($P = .014$), and macular edema ($P = .041$) significantly increased the incidence of depression. There was also a significant correlation between secondary glaucoma and anxiety ($P = .016$). In addition, patients who were current biologic agent users, IMT recipients, ever changed IMT regimens, or treated with frequent intraocular or periocular injections were significantly more likely to be depressed ($P = .059$, $P = .068$, $P = .039$, and $P = .003$, respectively). Notably, a higher VFQ composite score appeared protective against anxiety (OR = 0.954, 95% CI 0.953–0.975, $P < .0001$) and depression (OR = 0.94 95% CI 0.92–0.961, $P < .0001$).

Multivariate Analysis

Based on univariate analysis (Table 3), models were obtained by forward stepwise multiple regression analysis (Table 4). A higher annual household income or VFQ-25 composite score was protective against anxiety, while a poor BCVA in the better-seeing eye, discontinuing work due to uveitis, and uveitis-associated secondary glaucoma significantly increased the risk of anxiety. Current treatment with systemic biological agents, the presence of ocular complications, such as secondary glaucoma and complicated cataracts, and receiving frequent periocular or intraocular injections significantly increased the risk of depression. Higher socio-demographic characteristics, such as a higher educational level or annual household income, and higher VFQ-25 composite score were protective against depression. As shown in the Model HADS-A 3 and Model HADS-D 3, the VFQ-25 composite score was more predictive of anxiety and depression than BCVA, as the final models (Model HADS-A 3) comprised of the annual household income and VFQ-25 composite score explained nearly 74.4% of the variation in the HADS-A (OR = 0.398, 95% CI 0.174–0.911, $P = .029$ and OR = 0.958, 95% CI 0.938–0.978, $P < .0001$, respectively). The final model (Model HADS-D 3) comprised of ocular comorbidities and VFQ-25 composite score explained nearly 78.0% of the variation in HADS-D (OR = 2.259, 95% CI 1.2–4.252, $P = .012$ and OR = 0.943 95% CI 0.922–0.964, $P < .0001$, respectively). These values were much higher than those derived from other models (Figure 2).

Discussion

Compared with the control group, the QOL and mental health of patients with uveitis were significantly worse. Poor socio-demographic status, ocular complications, and poor self-reported visual function were predictors of mental health in patients with uveitis. VFQ-25 composite score was a more prominent predictor of anxiety and depression than BCVA. The best model consisted of low annual household income and poor self-reported visual function and accounted for 74.4% of the variation in HADS-A. The best model consisted of poor self-reported visual function and ocular complications and accounted for 78.0% of the variation in HADS-D. Patients with uveitis and psychological distress identified by these predictive indicators and models need to be included in routine uveitis clinical evaluations, as well as routine psychological clinical evaluations, to prevent adverse events and improve patient treatment compliance. This study emphasizes the need for attention toward the psychological problems and disease education of patients and suggests relevant departments to promote the improvement of the medical security system for patients with uveitis.

In this study, 26.5% of the uveitis group screened positive for depression based on the HADS, which is significantly higher than 9.5% in the reference group. This result is consistent with previous reports.[2,6] However, the incidence of depression in our patients was relatively low, which may be affected by different diseases and populations. Reasonable explanations for the high incidence of depression in patients with uveitis are the presence of chronic inflammation and cytokine dysregulation.[26,27] When the body experiences peripheral inflammation, the innate immune system produces inflammatory cytokines, including interleukin (IL)-1α, IL-1β, IL-6, and tumor necrosis factor-α, that can cross the blood–brain barrier and cause disease-associated issues, including

Table 2. Comorbidity and clinical characteristics of 245 patients with uveitis.

Variable	Total (N = 245) median (IQR) mean (SD) n (%)	Patients with uveitis grouped according to HADS-A score[a]			Patients with uveitis grouped according to HADS-D score[a]		
		Anxious (N = 41) median (IQR) mean (SD) n (%)	Not anxious (N = 204) median (IQR) mean (SD) n (%)	*P* value[b]	Depressed (N = 65) median (IQR) mean (SD) n (%)	Not depressed (N = 180) median (IQR) mean (SD) n (%)	*P* value[b]
Bilateral disease, (%)	140 (57.1%)	23 (56.1%)	117 (57.4%)	0.882	25 (38.5%)	80 (44.4%)	0.403
Median logMAR BCVA [c] in better-seeing eye, (IQR)	0 (0–0.10)	0 (0–0.10)	0 (0–0.07)	0.01	0.05 (0–0.22)	0 (0–0.05)	0.002[d]
Median logMAR BCVA [c] in worse-seeing eye, (IQR)	0.1 (0–0.3)	0.16 (0–0.57)	0.10 (0–0.36)	0.015	0.16 (0–0.52)	0.05 (0–0.3)	0.013[d]
Median duration (months), (IQR)	23 (6–52.2)	25 (6.00–71.00)	23 (6–51.23)	0.607	19.2 (7.1–66)	23.17 (6–51.3)	0.729[d]
Uveitis frequency, (%)				0.371			0.163
Onset	99 (40.4%)	14 (34.1%)	85 (41.7%)		31 (47.7%)	68 (37.8%)	
Recurrent	146 (59.6%)	27 (65.9%)	119 (58.3%)		34 (52.3%)	112 (62.2%)	
Location of inflammation, (%)				0.278			0.353
Anterior	129 (52.7%)	20 (48.8%)	109 (53.4%)		28 (43.1%)	101 (56.1%)	
Intermediate	22 (9.0%)	1 (2.4%)	21 (10.3%)		6 (9.2%)	16 (8.9%)	
Posterior	15 (6.1%)	3 (7.3%)	12 (5.9%)		4 (6.2%)	11 (6.1%)	
Panuveitis	75 (30.6%)	17 (41.5%)	58 (28.4%)		26 (40.0%)	49 (27.2%)	
Corneal uveitis or scleral uveitis	4 (1.6%)	0 (0%)	4 (2.0%)		1 (1.5%)	3 (1.7%)	
Diagnosis, (%)							
Viral infection uveitis	17 (6.9%)	2 (5.0%)	15 (7.3%)		5 (7.8%)	12 (6.7%)	
Behcet's disease	21 (8.6%)	7 (17.5%)	14 (7.0%)		10 (15.6%)	11 (6.2%)	
Vogt-Koyanagi-Harada syndrome	45 (18.4%)	9 (22.5%)	36 (17.9%)		12 (18.8%)	33 (18.6%)	
Ankylosing spondylitis associated AAU [c]	31 (12.7%)	7 (17.1%)	24 (11.8%)		5 (7.7%)	26 (14.4%)	
HLA-B27-related AAU [c]	23 (9.4%)	1 (2.5%)	22 (10.8%)		1 (1.6%)	22 (12.2%)	
Sympathetic ophthalmia	2 (0.8%)	0 (0%)	2 (1.0%)		1 (1.6%)	1 (0.6%)	
Fuchs' syndrome	14 (5.7%)	2 (5.0%)	12 (6.0%)		3 (4.7%)	11 (6.2%)	
Possner-Schlossman syndrome	20 (8.2%)	4 (10.0%)	16 (8.0%)		3 (4.7%)	17 (9.6%)	
Sarcoidosis	4 (1.6%)	0 (0%)	4 (2.0%)		1 (1.6%)	3 (1.7%)	
Sjögren syndrome associated uveitis	1 (0.4%)	0 (0%)	1 (0.5%)		0 (0%)	1 (0.6%)	
Idiopathic uveitis	67 (27.3%)	9 (22.0%)	58 (28.4%)		24 (36.9%)	43 (23.9%)	
Disease activity, (%)				0.797			0.564
Currently inactive	151 (61.6%)	26 (63.4%)	125 (61.3%)		42 (64.6%)	109 (60.6%)	
Active	94 (38.4%)	15 (36.6%)	79 (38.7%)		23 (35.4%)	71 (39.4%)	
Current systemic treatment							
Prednisolone, (%)	65 (26.5%)	13 (31.7%)	52 (25.5%)	0.411	22 (33.8%)	43 (23.9%)	0.119
Immunomodulatory therapy, (%)	52 (21.2%)	10 (19.2%)	42 (80.8%)	0.587	19 (29.2%)	33 (18.3%)	0.066
Single agent	20 (8.2%)	4 (9.8%)	16 (7.8%)		6 (9.2%)	14 (7.8%)	
Cyclosporine	11 (4.5%)	2 (4.9%)	9 (4.4%)		3 (4.6%)	8 (4.4%)	
Azathioprine	6 (2.4%)	2 (4.9%)	4 (2.0%)		2 (3.1%)	4 (2.2%)	
Methotrexate	1 (0.4%)	0 (0%)	1 (0.5%)		1 (1.5%)	0 (0%)	
Sulfasalazine	2 (0.8%)	0 (0%)	2 (1.0%)		0 (0%)	2 (1.1%)	
Combined agents	32 (13.1%)	6 (14.6%)	26 (12.7%)		13 (20%)	19 (10.6%)	
Cyclosporine & thalidomide	1 (0.4%)	0 (0%)	1 (0.5%)		1 (1.5%)	0 (0%)	
Cyclosporine & azathioprine	26 (10.6%)	5 (12.2%)	21 (10.3%)		10 (15.4%)	16 (8.9%)	
Cyclosporine & cyclophosphamide	2 (0.8%)	0 (0%)	2 (1.0%)		1 (1.5%)	1 (0.6%)	
Cyclosporine, azathioprine & cyclophosphamide	3 (1.2%)	1 (2.4%)	2 (1.0%)		1 (1.5%)	2 (1.1%)	
Biological agents, (%)	9 (3.7%)	3 (7.3%)	6 (2.9%)	.366	5 (7.7%)	4 (2.2%)	.104
History of changing, (%)							
Immunomodulatory therapy	37 (15.1%)	8 (19.5%)	29 (14.2%)	.387	15 (23.1%)	22 (12.2%)	.036
Ocular comorbidities, (%)							
None	143 (58.4%)	20 (48.8%)	123 (60.3%)	.172	26 (40.0%)	117 (65.0%)	.000
Cataract	51 (20.8%)	12 (29.3%)	39 (19.1%)	.144	20 (30.8%)	31 (17.2%)	.021
Glaucoma	31 (12.7%)	10 (24.4%)	21 (10.3%)	.013	14 (21.5%)	17 (9.4%)	.012
Vitreous hemorrhage	1 (0.4%)	0 (0%)	1 (0.5%)	1	0 (0%)	1 (0.6%)	1
Epiretinal membranes or vitreomacular traction syndrome	18 (7.3%)	4 (9.8%)	14 (6.9%)	.749	9 (13.8%)	9 (5.0%)	.039
Macular edema	25 (10.2%)	4 (9.8%)	21 (10.3%)	1	11 (16.9%)	14 (7.8%)	.037
Macular atrophy or Outer Retinal Disruption	10 (4.1%)	4 (9.8%)	6 (2.9%)	.114	5 (7.7%)	5 (2.8%)	.177
Retinal detachment with or without choroidal detachment	9 (3.7%)	3 (7.3%)	6 (2.9%)	.366	5 (7.7%)	4 (2.2%)	.104
Optic atrophy	4 (1.6%)	1 (2.4%)	3 (1.5%)	.522	1 (1.5%)	3 (1.7%)	1
Branch retinal vein occlusion or Branch retinal artery occlusion	2 (0.8%)	0 (0%)	2 (1.0%)	1	1 (1.5%)	1 (0.6%)	.461
Band keratopathy	2 (0.8%)	0 (0%)	2 (1.0%)	1	0 (0%)	2 (1.1%)	1
Choroidal neovascularization	1 (0.4%)	1 (2.4%)	0 (0%)	.167	1 (1.5%)	0 (0%)	.265
No light perception	1 (0.4%)	1 (2.4%)	0 (0%)	.167	1 (1.5%)	0 (0%)	.265
Retinal pigment epithelium detachment	1 (0.4%)	0 (0%)	1 (0.5%)	1	0 (0%)	1 (0.6%)	1
Periocular/intraocular injection, median, (IQR)	0 (0–2)	0 (0–2)	0 (0–2)	.903	1 (0–3)	0 (0–2)	.005[d]
Previous treatment regimen							
Prednisolone, (%)	104 (42.4%)	22 (53.7%)	82 (40.2%)	0.112	30 (46.2%)	74 (41.1%)	.481
Immunomodulatory therapy, (%)	67 (27.3%)	14 (34.1%)	53 (26.0%)	0.284	22 (33.8%)	45 (25.0%)	.17
Biological agents, (%)	33 (13.5%)	8 (19.5%)	25 (12.3%)	0.214	15 (23.1%)	18 (10.0%)	.008
Previous ocular surgery or laser therapy, (%)	68 (27.8%)	15 (36.6%)	53 (26.0%)	0.166	23 (35.4%)	45 (25.0%)	.109
Glaucoma surgery	13 (5.3%)	4 (9.8%)	9 (4.4%)		3 (4.6%)	10 (5.6%)	
Cataract surgery	46 (18.8%)	9 (22.0%)	37 (18.1%)		17 (26.2%)	29 (16.1%)	
Pars plana vitrectomy	5 (2.0%)	2 (4.9%)	3 (1.5%)		2 (3.1%)	3 (1.7%)	
Laser therapy[e]	35 (14.3%)	9 (22.0%)	26 (12.7%)		14 (21.5%)	21 (11.7%)	

[a]Chinese version of the Hospital Anxiety and Depression Scale: patients in the anxious and depressed group scored ≥8 and patients in the not anxious and not depressed group scored <8.

[b]Analyzed using a two-sided Chi-square test at a confidence level of 95%.

[c]BCVA: best-corrected visual acuity, AAU: acute anterior uveitis.

[d]Analyzed using a two-sided Wilcoxon rank-sum test at a confidence level of 95%.

[e]Laser therapy included laser peripheral iridotomy/selective laser trabeculoplasty (LPI/SLT) (18, 7.3%), YAG laser capsulotomy (11, 4.5%), retinal photocoagulation (9, 3.7%). One patient had undergone LPI/SLT and YAG laser capsulotomy, and two patients had undergone YAG laser capsulotomy and retinal photocoagulation.

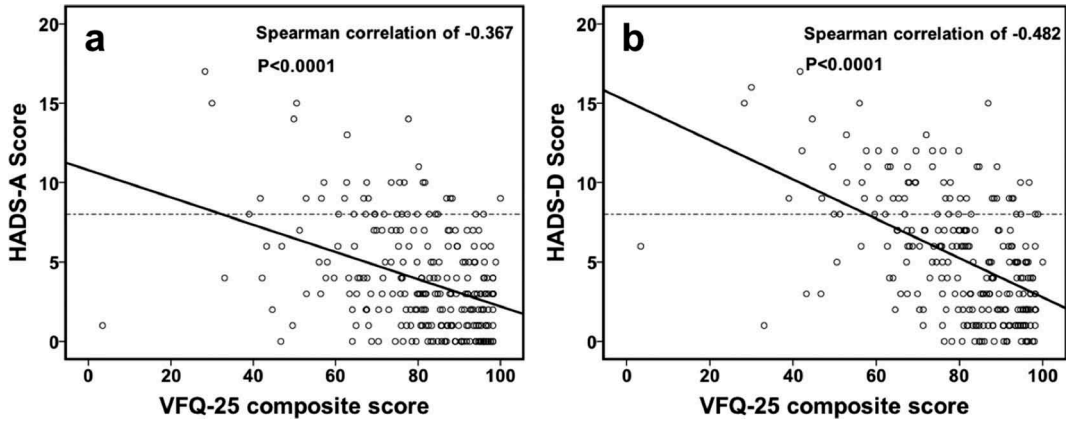


Figure 1. Correlation between CHI-HADS-Anxiety (HADS-A) score, CHI-HADS-Depression (HADS-D) score, and CHI-VFQ-25 composite score in patients with uveitis. (a) Correlation between HADS-A score and CHI-VFQ-25 composite score in patients with uveitis (with a Spearman correlation of −0.367, $P < .0001$). (b) Correlation between HADS-D score and CHI-VFQ-25 composite score in patients with uveitis (with a Spearman correlation of −0.482, $P < .0001$). Dashed line represents the cutoff score of 8 of HADS-A and HADS-D between the negative and positive screen.

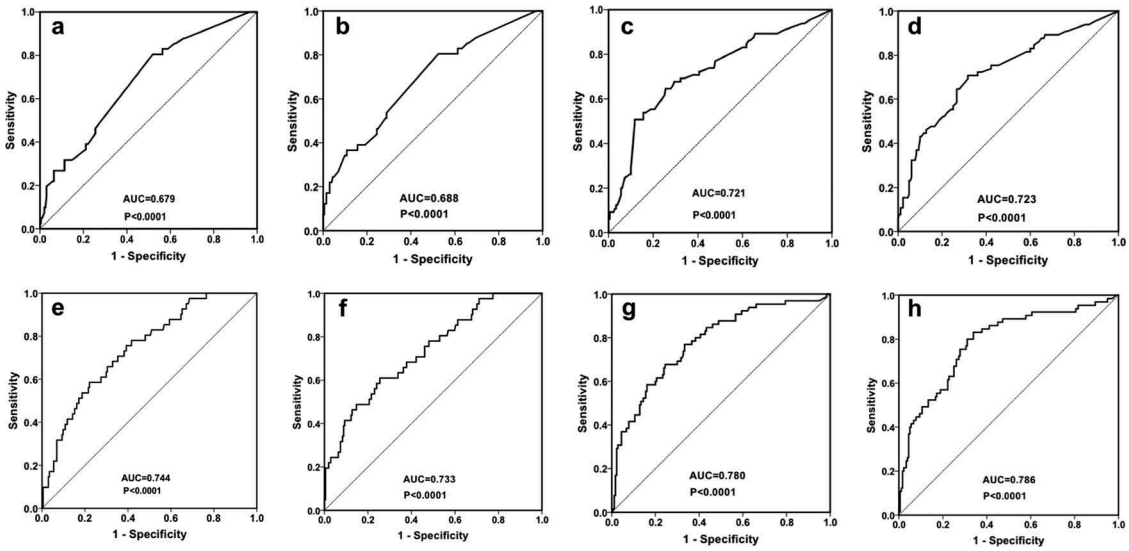


Figure 2. Receiver operating characteristic (ROC) curve analyses.(a, b, e, f) Receiver operating characteristic (ROC) curve and the corresponding area under the curve (AUC) of (a) Model-HADS-A 1, (b) Model-HADS-A 2, (e) Model-HADS-A 3 and (f) Model-HADS-A 4. (c, d, g, h) are the ROC curve and the corresponding AUC of (c) Model-HADS-D1, (d) Model-HADS-D 2, (g) Model-HADS-D 3 and (h) Model-HADS-D 4.

depression.[26,28] In addition, systemic or intraventricular injection of cytokines has been found to cause depression-like behavior in mice.[29]

Lower education levels, discontinuing work due to uveitis, and lower household income are risk factors for anxiety and depression. These results are consistent with a previous study showing psychological distress in old age caused by financial hardship in childhood may be reduced by education and employment later in life.[30] The protective mechanism of higher education may involve personal characteristics associated with varying levels of stress tolerance and different occupational restrictions, which may be related to stress exposure levels and physical health.[31] Discontinuing work adds to the economic burden of individuals and causes mental illness. Depression and suicidal behavior accompany economic crises in low-income and middle-income countries, while some affluent countries offer provisions to help the unemployed escape adverse consequences.[32] However, the reimbursement ratio of the medical insurance of these patients needing long-term clinical care is relatively low in China.[33] Our results remind clinicians to consider the patient's financial situation when considering treatment. In addition, authorities need to improve the medical security system for patients with chronic inflammatory diseases that require long-term or even lifelong medication.

The current use of biologic agents is a risk factor for depression. The use of interferon-α, a type of biologic agent systemically used to treat ocular Behçet disease, has been associated with increased depression and suicidal ideation in patients.[34] However, a previous study found anti-tumor necrosis factor has a therapeutic effect on patients with major depression and C-reactive protein upregulation.[35] The possible reason for the result of our study may be that biologics are still fully self-funded in China, but are covered by medical insurance in some western countries. Due to their high price and individual differences in therapeutic effects, the

Table 3. Univariate logistic regression analysis of anxiety and depression[a] in patients with uveitis.

	Anxiety[a]		Depression[a]	
Variable	OR (95% CI)-crude	*P* value[b]	OR (95% CI)-crude	*P* value[b]
Educational level	0.594 (0.364 to 0.97)	.037	0.52 (0.341 to 0.793)	.002
Discontinuing work due to uveitis	3.879 (1.474 to 10.21)	.006	2.469 (0.973 to 6.266)	.057
Annual household income	0.329 (0.15 to 0.72)	.006	0.428 (0.231 to 0.793)	.007
LogMAR BCVA[c] in the better-seeing eye	4.157 (1.212 to 14.256)	.023	4.651 (1.24 to 17.446)	.023
LogMAR BCVA[c] in the worse-seeing eye	1.933 (1.073 to 3.482)	.028	1.814 (1.063 to 3.098)	.029
VFQ-25 composite score	0.954 (0.935 to 0.975)	0	0.94 (0.92 to 0.961)	0
Current systemic biological agent therapy	‡		3.667 (0.953 to 14.102)	.059
Ocular comorbidities	‡		2.786 (1.555 to 4.991)	.001
Complicated cataract	‡		2.136 (1.111 to 4.107)	.023
Secondary glaucoma	2.811 (1.209 to 6.536)	.016	2.632 (1.214 to 5.708)	.014
Macular edema	‡		2.415 (1.035 to 5.636)	.041
Current systemic IMT[c]	‡		1.84 (0.956 to 3.54)	.068
History of changing IMT[c]	‡		2.155 (1.039 to 4.468)	.039
Periocular/intraocular injection	‡		1.196 (1.061 to 1.347)	.003

OR = odds ratio, CI = confidence interval.

[a]Chinese version of the Hospital Anxiety and Depression Scale: patients in the anxious and depressed group scored ≥8 and patients in the not anxious and not depressed group scored <8.

[b]Univariate logistic regression analysis at a confidence level of 95%.

[c]BCVA: best-corrected visual acuity, IMT: immunomodulatory therapy.

[‡]This variable was not statistically significant in univariate logistic regression analysis.

Variables that might have clinical and sociological significance, such as discontinuing work due to uveitis, current systemic biological agent therapy, and current systemic IMT were included in this analysis.

Table 4. Multivariate logistic regression model[a] predicting anxiety and depression[b] in patients with uveitis.

Variable	Model HADS-A 1 [c] OR (95%CI)-adj	Model HADS-A 2 [c] OR (95%CI)-adj	Model HADS-A 3 [c] OR (95%CI)-adj	Model HADS-A 4 [c] OR (95%CI)-adj	Model HADS-D 1[d] OR (95%CI)-adj	Model HADS-D 2[d] OR (95%CI)-adj	Model HADS-D 3[d] OR (95%CI)-adj	Model HADS-D 4[d] OR (95%CI)-adj
Education level					0.663 (0.416–1.055)	0.633 (0.396–1.013)		
Discontinuing work due to uveitis	2.473 (0.884–6.919)	2.333 (0.821–6.632)						0.536 (0.269–1.07)
Annual household income	0.356 (0.158–0.805) *	0.372 (0.164–0.843) *	0.398 (0.174–0.911) *	0.413 (0.18–0.948) *	0.536 (0.273–1.052)	0.521 (0.264–1.026)		
LogMAR BCVA[e] in the better-seeing eye	3.25 (0.898–11.769)	2.948 (0.786–11.055)						
LogMAR BCVA[e] in the worse-seeing eye								
VFQ-25 composite score			0.958 (0.938–0.978) ****	0.958 (0.938–0.979) ****			0.943 (0.922–0.964) ****	0.946 (0.925–0.968) ****
Current systemic biological agent therapy					4.81 (1.136–20.359) *	4.848 (1.16–20.254) *		
Ocular comorbidities					2.143 (1.144–4.013) *		2.259 (1.2–4.252) *	
Complicated cataract						1.821 (0.896–3.7)		
Secondary glaucoma		2.230 (0.905–5.491)		2.251 (1.005–6.254) *		2.439 (1.117–5.775) *		2.623 (1.116–6.166) *
Macular edema								
Current systemic IMT[e]								
History of changing IMT[e]								
Periocular/intraocular injection					1.154 (1.02–1.305) *	1.179 (1.042–1.334) **		1.151 (1.01–1.311) *
AUC[e] of ROC[e] (95% CI)	0.679 (0.591–0.767) ****	0.688 (0.597–0.779) ****	0.744 (0.667–0.822) ****	0.733 (0.650–0.816) ****	0.721 (0.645–0.797) ****	0.723 (0.648–0.799) ****	0.780 ((0.714–0.847) ****	0.786 (0.719–0.853) ****

OR = odds ratio, CI = confidence interval

[a]Multivariate logistic regression model: α entry = 0.10, α removal = 0.15. *P*: * <0.05, **<0.01, ***<0.001, ****<0.0001.

[b]Chinese version of the Hospital Anxiety and Depression Scale: patients in the anxious and depressed group scored ≥8 and patients in the not anxious and not depressed group scored <8.

[c]Models predicting anxiety. Variables included in the multivariate regression analysis: 1) HADS-A 1: education level, discontinuing work due to uveitis, annual household income, BCVA in the better-seeing eye (logMAR), BCVA in the worse-seeing eye (logMAR); 2) HADS-A 2: education level, discontinuing work due to uveitis, annual household income, BCVA in the better-seeing eye (logMAR), BCVA in the worse-seeing eye (logMAR), secondary glaucoma; 3) HADS-A 3: education level, discontinuing work due to uveitis, annual household income, VFQ-25 composite score; 4) HADS-A 4: education level, discontinuing work due to uveitis, annual household income, VFQ-25 composite score, secondary glaucoma.

[d]Models predicting depression. Variables included in the multivariate regression analysis: 1) HADS-D 1: education level, discontinuing work due to uveitis, annual household income, BCVA in the better-seeing eye (logMAR), BCVA in the worse-seeing eye (logMAR), current systemic biologic agent treatment, ocular comorbidities, current systemic IMT, history of changing IMT, periocular/intraocular injection times; 2) HADS-D 2: education level, discontinuing work due to uveitis, annual household income, BCVA in the better-seeing eye (logMAR), BCVA in the worse-seeing eye (logMAR), current systemic biologic agent treatment, complicated cataract, secondary glaucoma, macular edema, current systemic IMT, history of changing IMT, periocular/intraocular injection times; 3) HADS-D 3: education level, discontinuing work due to uveitis, annual household income, VFQ-25 composite score, current systemic biologic agent treatment, ocular comorbidities, current systemic IMT, history of changing IMT, periocular/intraocular injection times; 4) HADS-D 4: education level, discontinuing work due to uveitis, annual household income, VFQ-25 composite score, current systemic biologic agent treatment, complicated cataract, secondary glaucoma, macular edema, current systemic IMT, history of changing IMT, periocular/intraocular injection times.

[e]BCVA: best-corrected visual acuity, IMT: immunomodulatory therapy, ROC: receiver operating characteristic curve, AUC: area under curve.

use of biologic agents sometimes aggravates the economic and mental burden of patients. Furthermore, biologic agents are usually used in patients with severe disease who have failed to respond to systemic hormone therapy and IMT. In this context, the addition of biological agents may mean an increased difficulty in curing the disease, resulting in increased psychological burden. Also, the majority of patients treated with biologic agents had poor visual function and there is a clear relationship between poor visual function and depression. Moreover, the application of biologic agents in uveitis is not common in China and patients lack awareness of and receptivity to these new drugs.

No correlation was found between current oral corticosteroid therapy or IMT and depression, yet studies have shown IMT can be beneficial to mental health.[36] Notably, univariate analysis showed a history of changing IMT was predictive of depression, which is consistent with findings by Ying Qian et al.,[37] perhaps indicating that patients with more complex and intractable diseases have a higher risk of depression. However, this factor is no longer significant following multivariate regression analysis. The median duration (IQR) of diseases of all the patients was 23 (6–52.2) months, no statistically significant correlation between the duration of disease (P = .729) and depression was found. There was also no significant correlation between disease activity and depression (P = .564). A possible explanation is that a cross-sectional study can have difficulties in consistency of dynamic disease capture.

This research indicates secondary glaucoma associated with uveitis is a risk factor for anxiety and depression. A large sample of 4,439,518 patients revealed a statistically significant correlation between glaucoma and anxiety and depression with an adjusted ORs of 10.6 (95%CI 10.0–11.0) and 12.3 (95% CI 11.8–12.9), respectively.[17] Our study also linked complicated cataracts and depression. In the elderly, the correlation between cataract-related visual impairment and depression was significantly independent of demographics, life exposure, and current vision.[38–40] After surgery, the VRQOL of elderly patients significantly improves, which is followed with a decrease in depression and cognitive impairment.[41] This may aid clinicians in determining the timing of surgery. However, this conclusion cannot be easily extended to patients with uveitis complicated with cataract, because the average age of patients with uveitis complicated with cataract is different from that of patients with cataract, there were age-related difference in cognition between the two groups. Univariate logistic regression analysis found that there was a certain correlation between macular edema and depression (P = .041), while the significance was lost in the multivariate model. The incidence of uveitic macular edema (10.2%) in this study was lower than that in another study,[42] possibly because this was a cross-sectional study, a certain number of patients were in recovery. Therefore, the present study was of limitation in suggesting a correlation between macular edema and depression. Research has shown that diabetic macular edema was not associated with depressive and anxious symptoms.[43] To our knowledge, there have been no relevant studies focusing on anxiety and depression in patients with these three complications of uveitis. Further prospective longitudinal research on the mental health status of these groups may provide a new theoretical basis for the selection of surgical timing and rational treatment.

This study found the risk of depression increased as the number of periocular or intraocular injections increased. There are a few possible explanations. Patients' confidence declined as the number of injections increased. In fact, most patients who were treated more frequently were also those experiencing frequent relapses. Additionally, frequent local injection therapy increases the incidence of local complications. Previous study has shown that patients being treated with systemic anti-inflammatory therapy did not have significant systemic side effects, while patients treated with fluocinolone acetonide intraocular implants had a higher incidence of cataract and secondary glaucoma.[44] Our results suggest clinicians should reasonably choose systematic or local treatments based on the psychological status of patients with uveitis.

Studies have linked visual impairment to mental illness.[45] Our study also showed that poor BCVA in the better-seeing eye was a risk factor for positive-HADA, which is consistent with previous research results.[9] This is possibly because BCVA in the better-seeing eye determines a person's visual potential. However, Spearman correlation analysis identified a mild or moderate correlation between VFQ composite score and HADS-A and HADS-D score, possibly because this was a cross-sectional study involving a non-single disease with patients at different stages of treatment. The previous study has found a significant correlation between BCVA and VRQOL.[46] This study attempted to explore which of the above two factors has a stronger role in indicating the risk of anxiety and depression. Multiple regression analysis revealed the VFQ-25 composite score was a more prominent predictor of positive-HADA and positive-HADD than BCVA, and the AUC of the ROC of the final regression model notably increased to 0.744 and 0.780, respectively, when the VFQ-25 composite score was considered. A similar phenomenon has been observed in patients with ocular inflammatory disease.[37] Visual function was not only reflected in vision but also in contrast sensitivity and visual field, which were also associated with mental problems.[18] Furthermore, patient's visual functioning is important in all aspects of life, including activities near and far, self-care, social contact, and mental health. Thus, it is not realistic to discuss visual function separately from specific life situations. VFQ-25, containing 12 dimensions, can more comprehensively evaluate patient visual function and VRQOL than BCVA. Depressed patients scored lower in almost all dimensions of the VFQ-25 than non-depressed patients. Therefore, BCVA is not a sufficient indicator with which to assess patient vision and the contribution of VRQOL scales, like VFQ-25, is indispensable.

Based on the above research results, it is of great significance to evaluate the QOL of patients reasonably. Apart from the indirect method of questionnaires, the direct methods also

play an important role in QOL assessment, because they are more sensitive and comprehensive. Time trade-off (TTO) and standard gamble (SG), commonly used direct methods, allow QOL to be compared among different health conditions without the need for mapping.[47,48] For example, by comparing the TTO utility value, researchers found that the HRQOL in patients with uveitis with poor vision was worse than that of patients with AIDS.[48,49] However, TTO and SG presented cognitive bias for some patients,[50] and sometimes, the delivery of questions related to death may cause discomfort and panic in subjects. In contrast, questionnaires are generally easier to answer, and patients may be more willing to cooperate.[48] Therefore, the indirect and direct methods have their own advantages and disadvantages, and rational development and application of the above two methods will help to objectively evaluate the QOL of patients.

Based on our results, the psychological status of patients with uveitis should be concerned. Several studies have shown that anxiety and depression can affect the treatment compliance of patients with some chronic diseases, thus affecting treatment efficacy.[51] Wantanee Sittivarakul attributed poor understanding of disease by patients with ocular inflammatory disease to the cognitive impairment complicated by anxiety and depression.[52] A similar association has been found in patients with glaucoma.[53] Due to the higher tendency for depression, clinicians should pay more attention to the disease education of patients with uveitis. In addition, early detection and timely intervention in the mental distress of patients with uveitis may be of great significance in restoring cognition and improving patient compliance.

There are several limitations in our study. First, the enrolled patients are not representative of all patients with uveitis in China, so the results may not be generalizable to other populations. Second, while the HADS scale used in the study has been verified to have good sensitivity and specificity at the threshold of 8, it is still not a formal psychiatric evaluation for the diagnosis of anxiety and depression. Third, since our study was a cross-sectional study, a significant number of our patients were in the disease stabilization and recovery stage, so visual dysfunction and symptoms of anxiety and depression may be lower in our patient population than in other clinical studies. In addition, considering systemic corticosteroid and immunomodulatory treatment, dosage and duration of administration were not considered as stratified variables, which may inhibit assessing the relationships between systemic therapy and anxiety and depression. Therefore, longitudinal studies to determine differences in QOL and psychological status of patients at different stages of treatment are important avenues of future study.

In this study, we evaluated the impact of uveitis on the QOL and mental health of patients. Low socio-demographic characteristics, ocular complications, and poor self-reported visual function were identified as important factors when predicting the mental health of patients with uveitis. Compared with the BCVA, self-reported visual function was a better predictor of mental problems. Our findings provide support for heightened attention and awareness toward psychological distress in patients with uveitis, as well as remind clinicians of the significance of disease education. Furthermore, perfecting the medical guarantee system is particularly important for patients with uveitis.

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Declaration of interest

The authors report no conflicts of interest. The authors alone are responsible for the content and writing of the paper.

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Contributors

YW, YJ: conception and design. YJ, MJ, RL,: execution of the study. YJ, JZ, YMY: analysis and interpretation of data. YJ, DL: drafting the article. YW, YJ, XFH, MLD, YLW, XFZ: revising it critically for important intellectual content. YW, YJ, XFH: final approval of the version to be published.

Ethics approval

Review committee of Wenzhou Medical University.

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